DEEP LEARNING EN SISTEMAS RECOMENDADORES Y FILTRADO COLABORATIVO

23 de Septiembre, 2025

EN ESTA CLASE

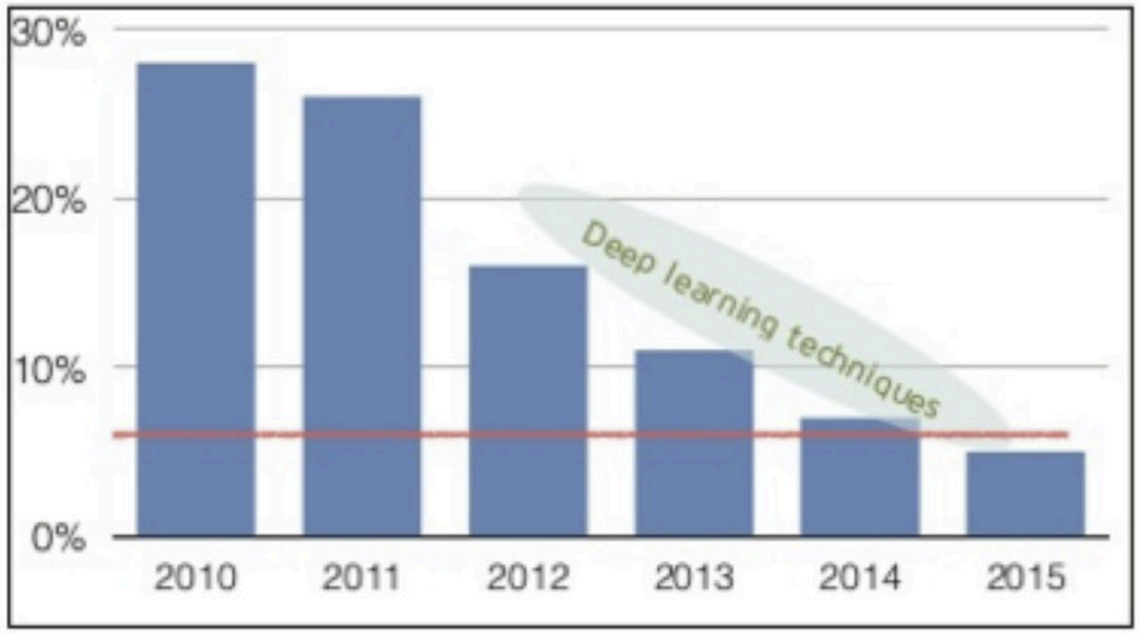
01

INTRODUCCIÓN A REDES NEURONALES

02

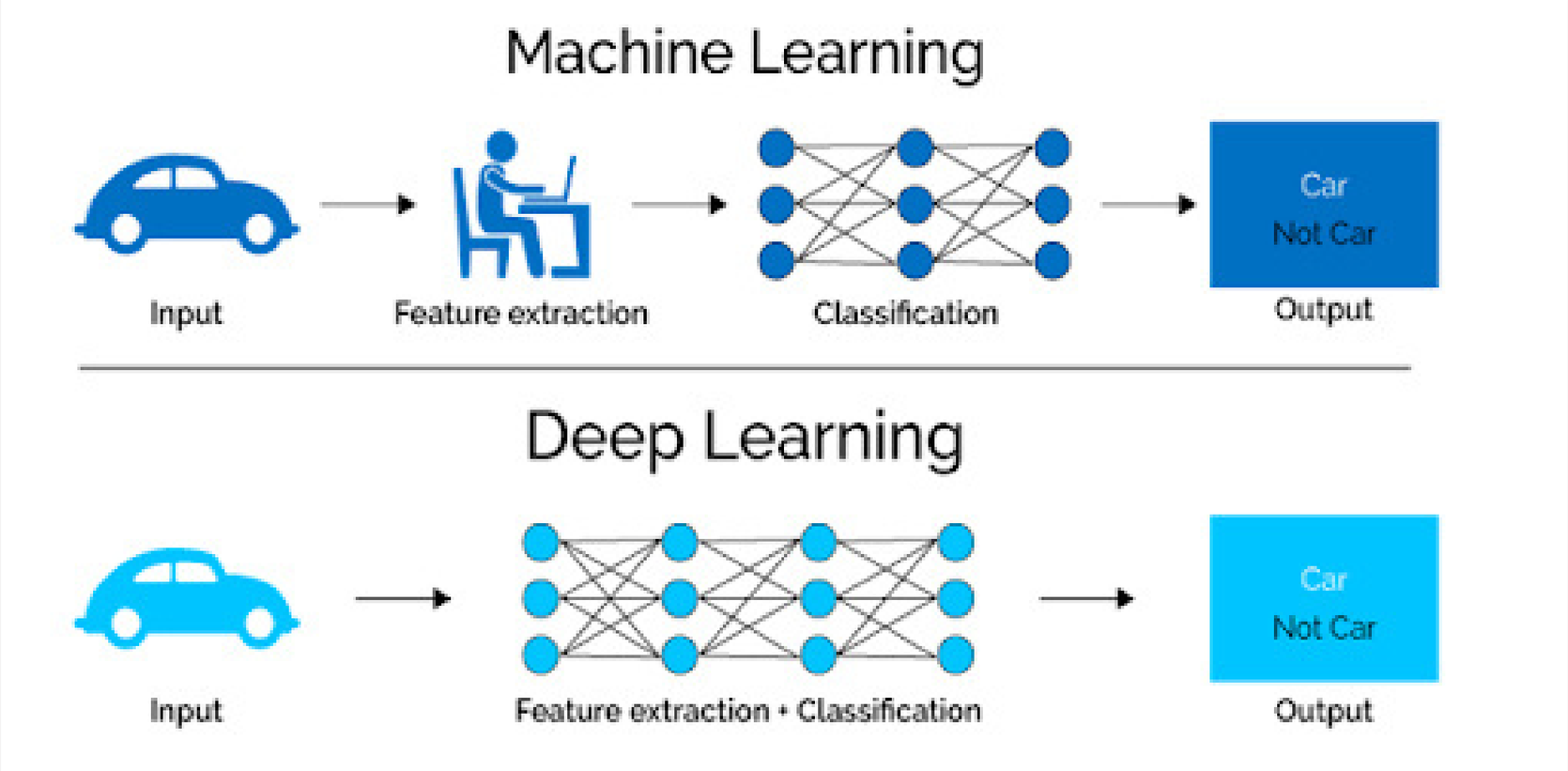
RECOMENDACIÓN CON REDES NEURONALES

¿POR QUÉ DEEP LEARNING?

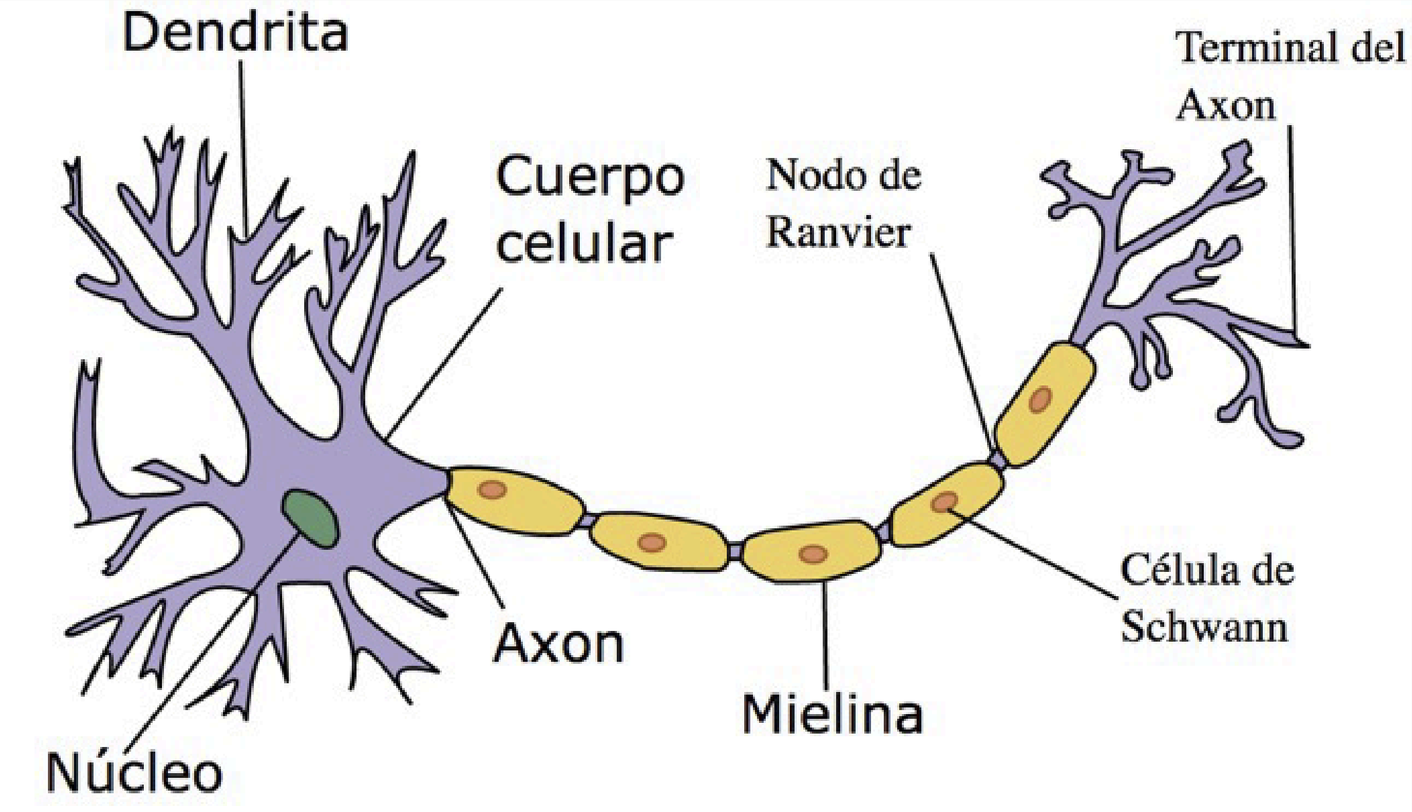


ImageNet challenge error rates (red line = human performance)

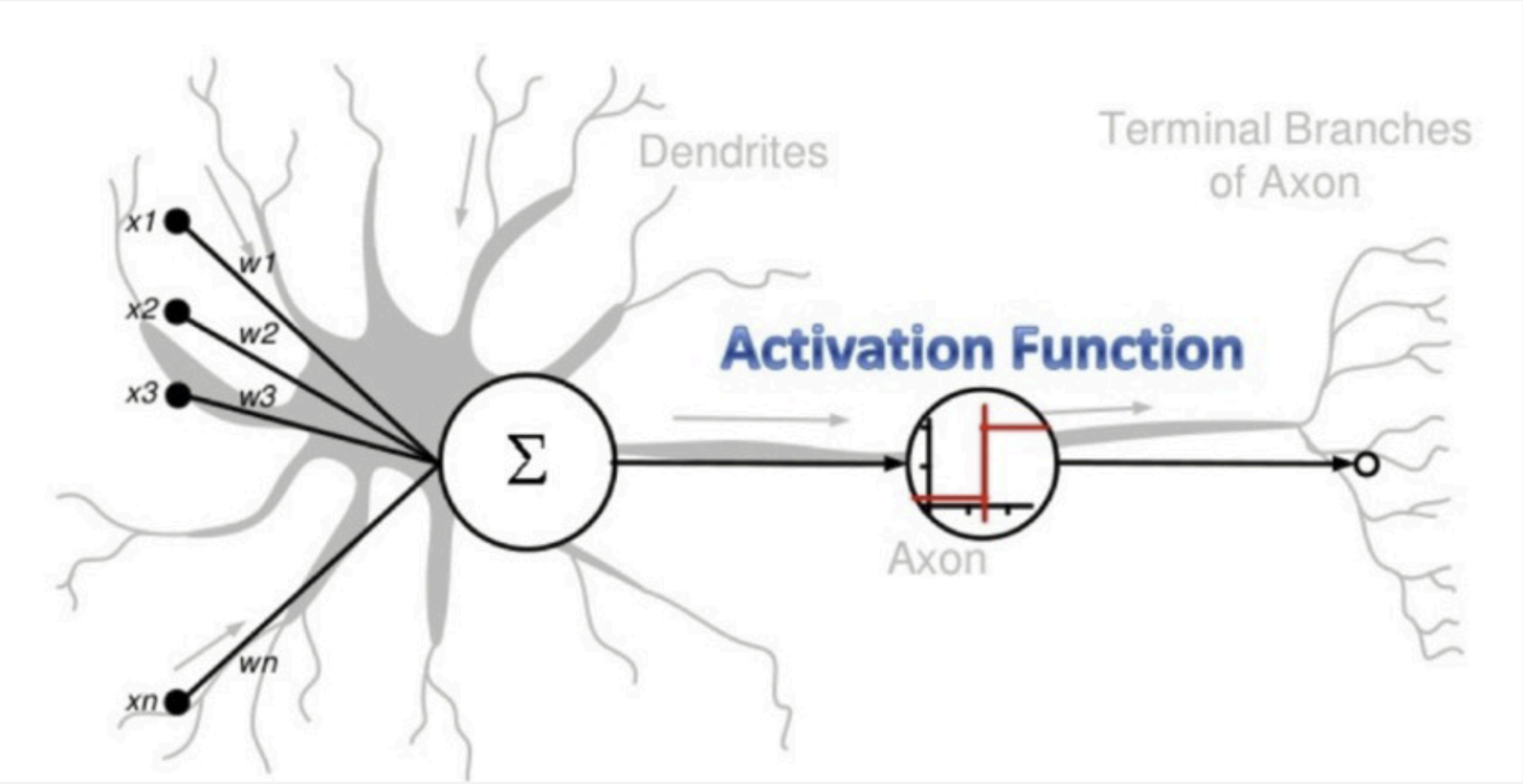
¿POR QUÉ DEEP LEARNING?



BASE BIOLÓGICA



BASE BIOLÓGICA

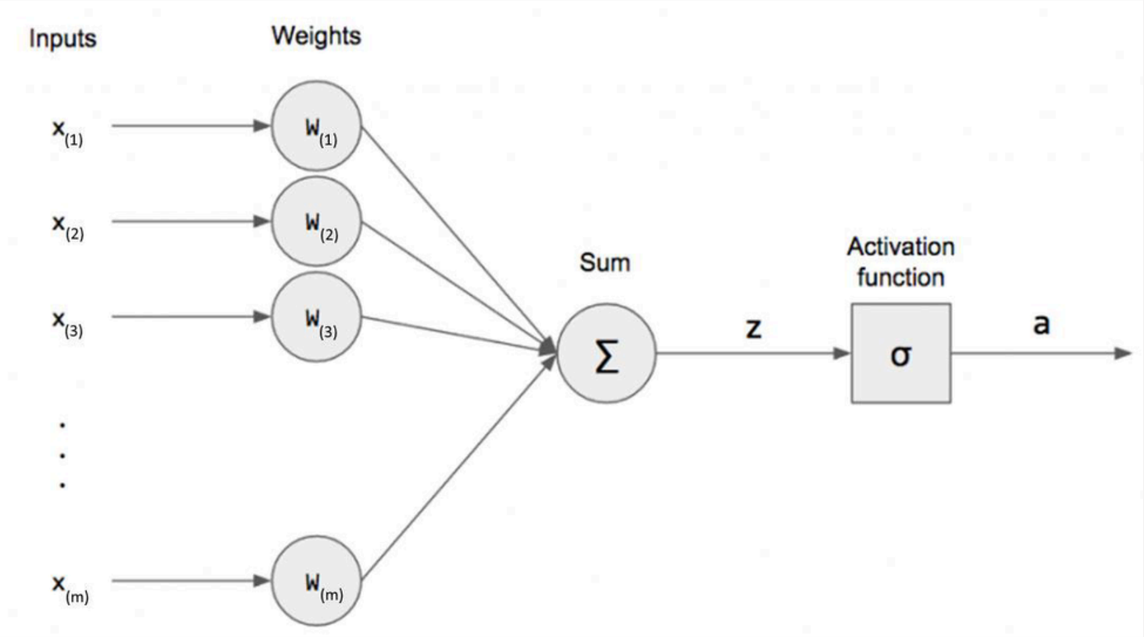


PERCEPTRÓN

- Propuesto en 1957 por Frank Rosenblatt

  “[The Perceptron is] the embryo of an electronic computer that [the Navy] expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence”

  The New York Times

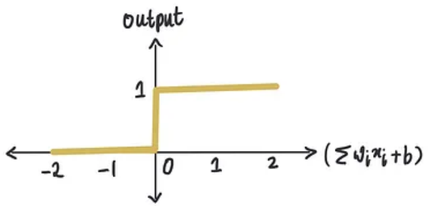


FUNCIONES DE ACTIVACIÓN

binary step

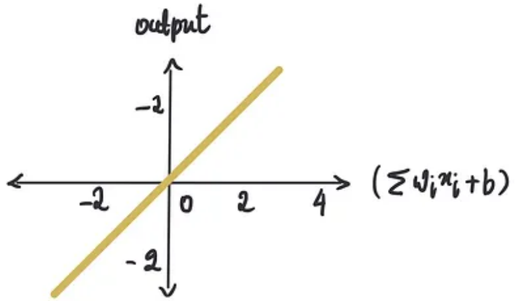
$$f(x) = \begin{cases} 0 & , x<0 \\ 1 & , x \geq 0 \end{cases}$$

Some input



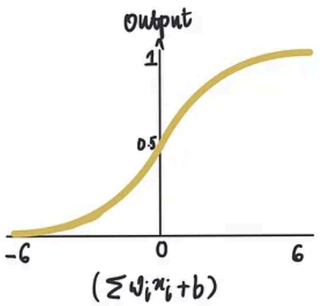
linear activation

$$f(x) = x$$



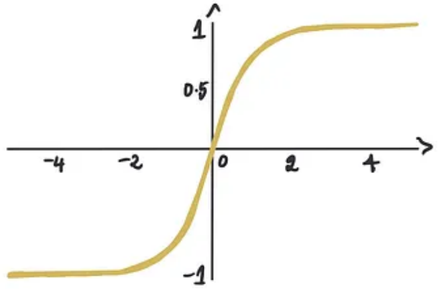
sigmoid

$$f(x) = \frac{1}{1+e^{-x}}$$



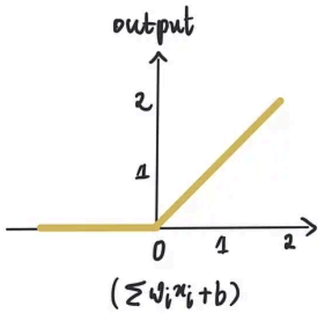
hyperbolic tangent

$$f(x) = \frac{1 - e^{-2x}}{1 + e^{-2x}}$$



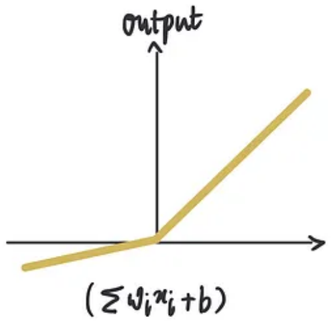
rectified linear units

$$f(x) = \max(x, 0)$$



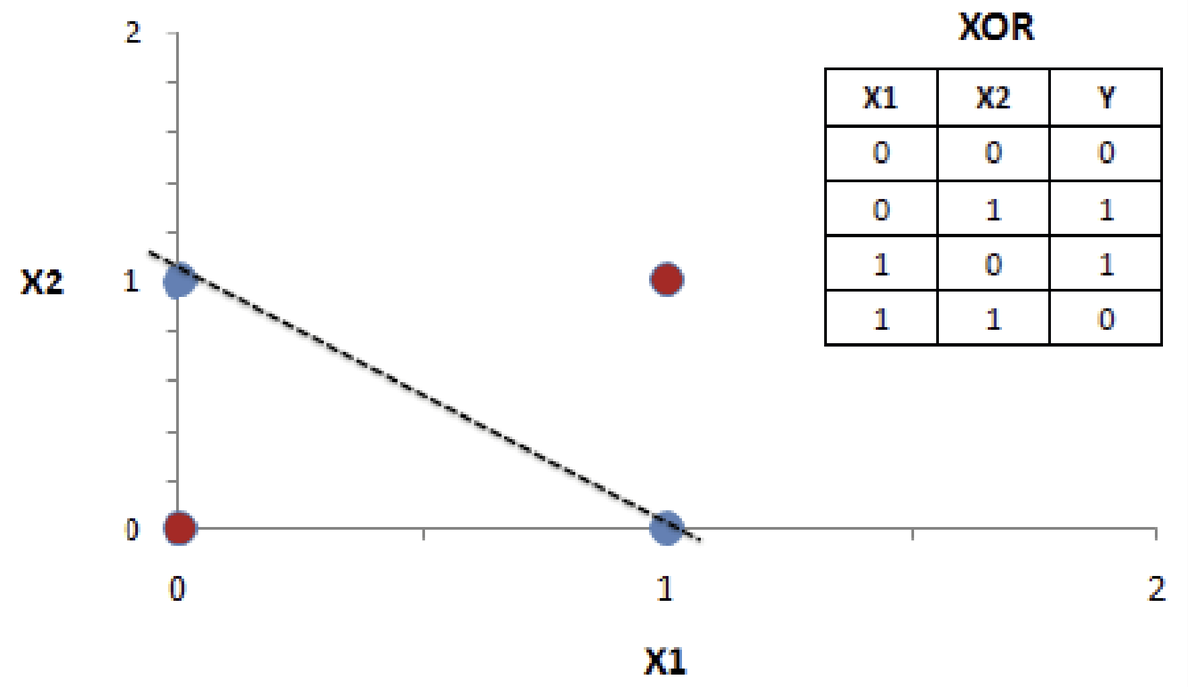
leaky ReLU

$$f(x) = \begin{cases} 0.01x & \text{for } x<0 \\ x & \text{for } x \geq 0 \end{cases}$$

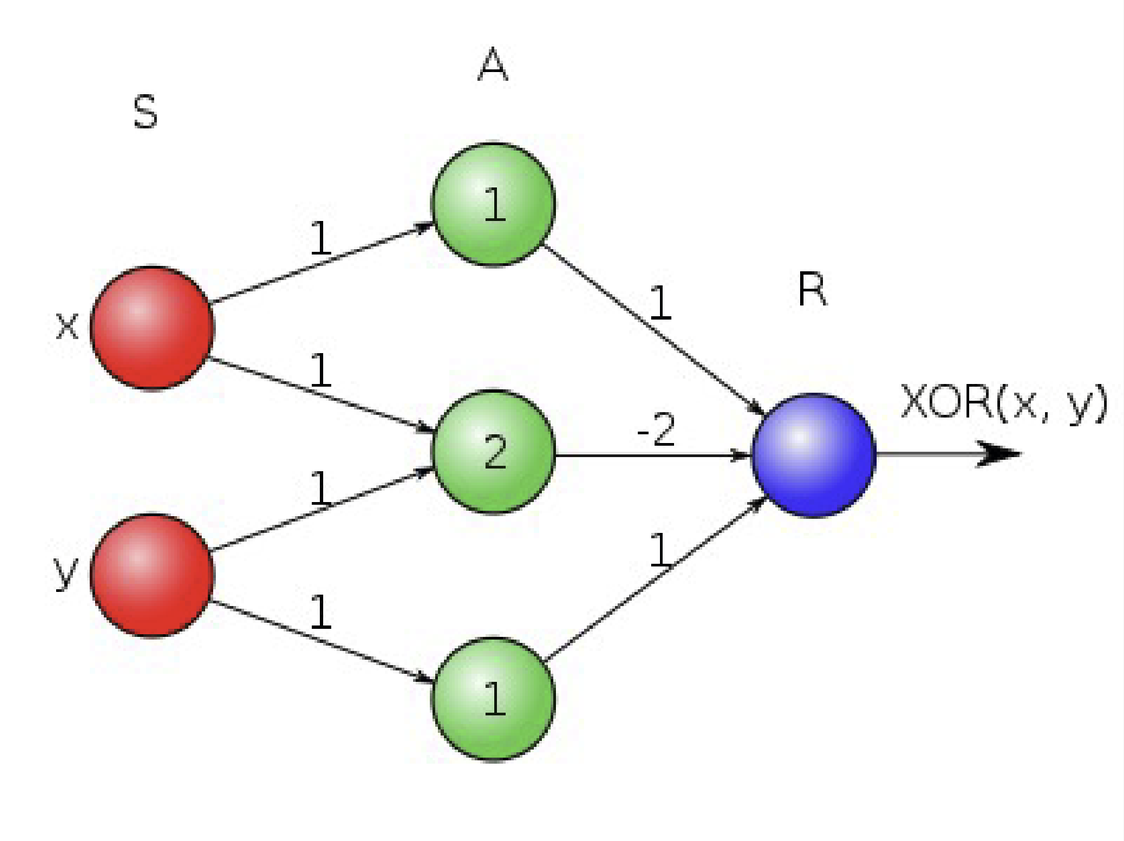


LIMITACIONES DEL PERCEPTRÓN

- El perceptrón de una capa es un clasificador lineal
- No puede separar algunas funciones como XOR



PERCEPTRÓN MULTICAPA

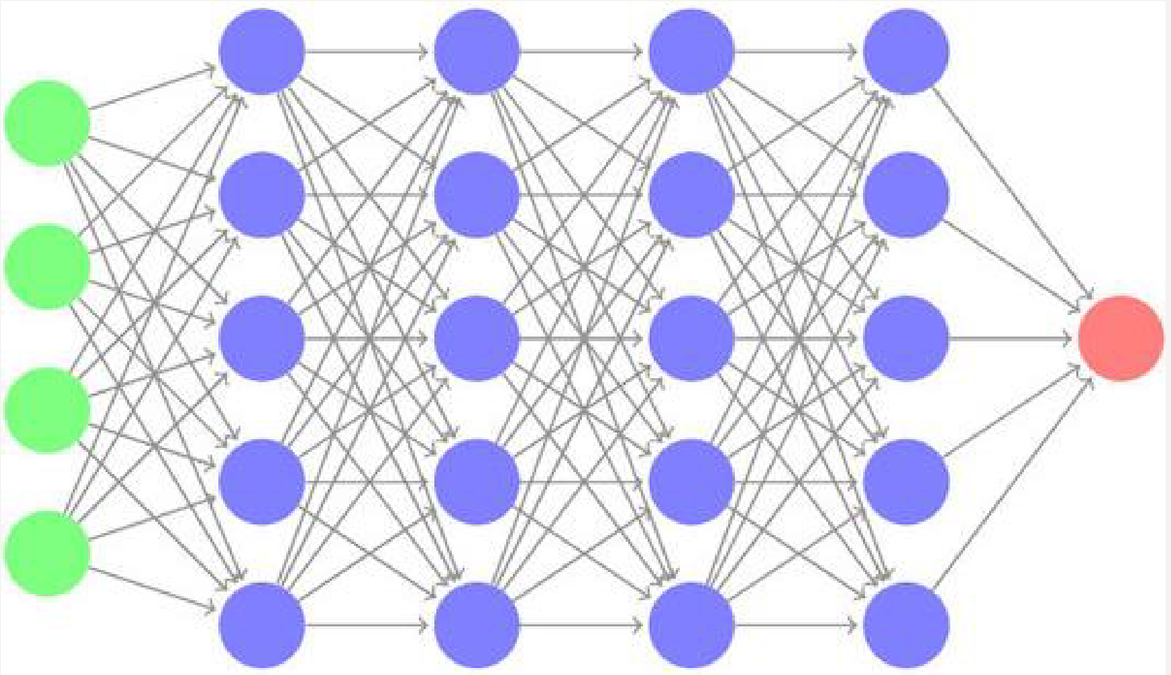


REDES FEEDFORWARD MULTILAYER

$$h^{(0)} = X$$

$$h^{(i)} = f^{(i)}(h^{(i-1)}W^{(i)} + b^{(i)})$$

$$y = g(h^{(L)}U + c)$$



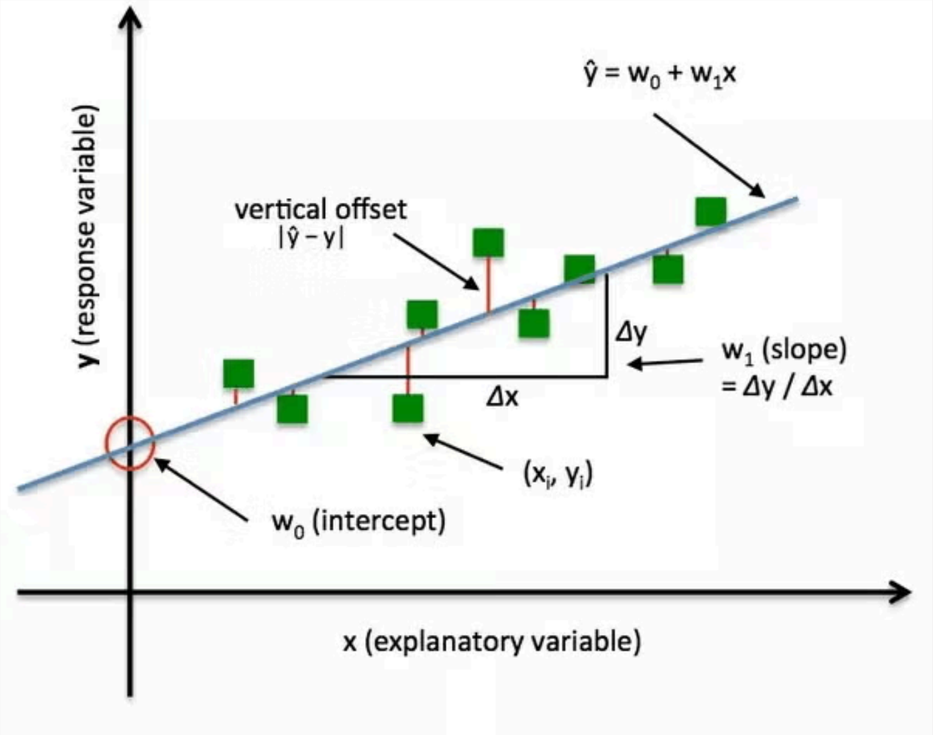
FUNCIONES DE PÉRDIDA

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$\text{BCE} = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right]$$

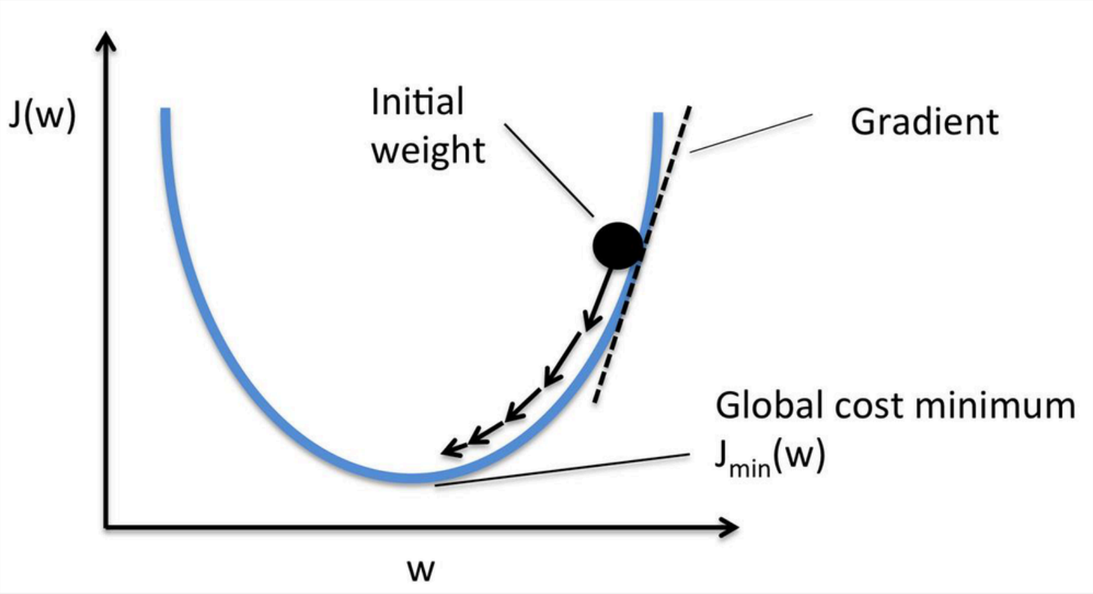
$$\text{CCE} = -\sum_{i=1}^{n}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c})$$



DESCENSO DE GRADIENTE

Para encontrar los parámetros óptimos utilizamos Descenso de Gradiente:

$$\hat{\theta} = \hat{\theta} - \lambda \frac{\partial L}{\partial \theta}\bigg|_{\hat{\theta}}$$

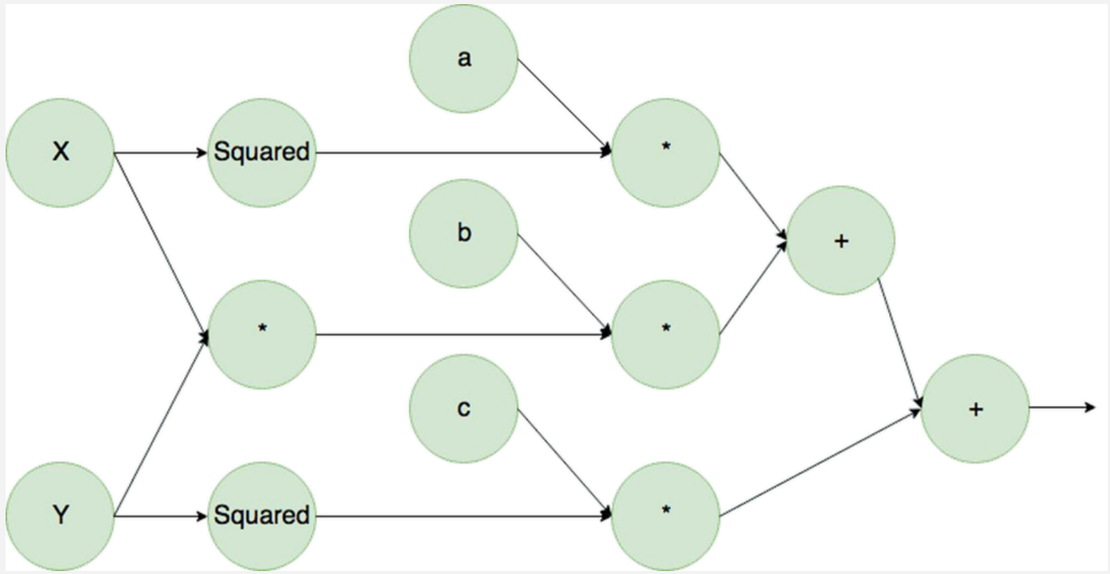


BACKPROPAGATION

Es una técnica que utiliza la Regla de la Cadena y Programación Dinámica para encontrar de manera eficiente el valor de las derivadas

$$y = f(x) \quad x = g(u)$$

$$\frac{dy}{du} = \frac{dy}{dx} \cdot \frac{dx}{du}$$



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SLIM

- SLIM: Sparse Linear Methods
- Se basa en modelos lineales dispersos.
- Trabaja en base a una matriz de interacciones para realizar recomendaciones Top-N
- Aprende una matriz sparse W la influencia de otros ítems en el interés de un ítem en particular

$$\min_{W} \ \frac{1}{2}\|A - AW\|_F^2 + \frac{\beta}{2}\|W\|_F^2 + \lambda\|W\|_1$$

$$\text{subject to } W \geq 0,\ \text{diag}(W) = 0$$

$$\tilde{A} = AW$$

X. Ning and G. Karypis, "SLIM: Sparse Linear Methods for Top-N Recommender Systems," 2011 IEEE 11th International Conference on Data Mining, Vancouver, BC, Canada, 2011, pp. 497-506

https://github.com/KarypisLab/SLIM

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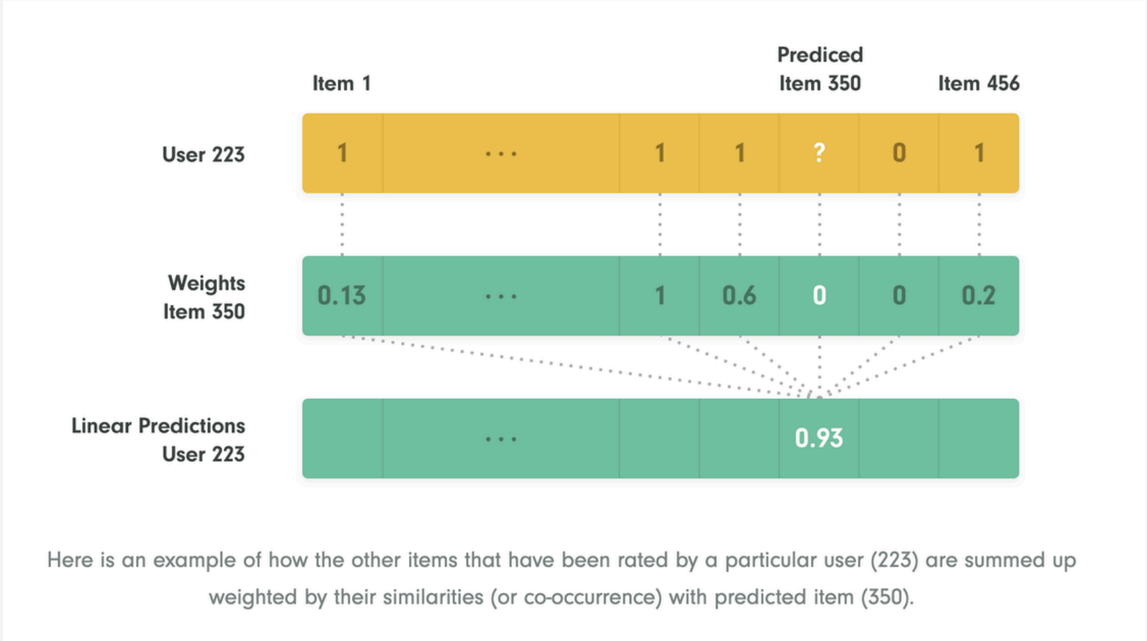

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EASE

- EASE: Embarrassingly Shallow AutoEnconders (R)
- Se inspira en autoencoders, pero de manera simplificada al no usar capas ocultas
- Trabaja en base a una matriz de interacciones para realizar recomendaciones Top-N
- Tiene solución en forma cerrada

$$\min_B \|X - XB\|_F^2 + \lambda \|B\|_F^2$$

$$\text{subject to } \operatorname{diag}(B) = 0$$

$$S_{uj} = X_{u,\cdot} \cdot B_{\cdot,j}$$

Steck, H. (2019, May). Embarrassingly shallow autoencoders for sparse data. In The World Wide Web Conference (pp. 3251-3257).

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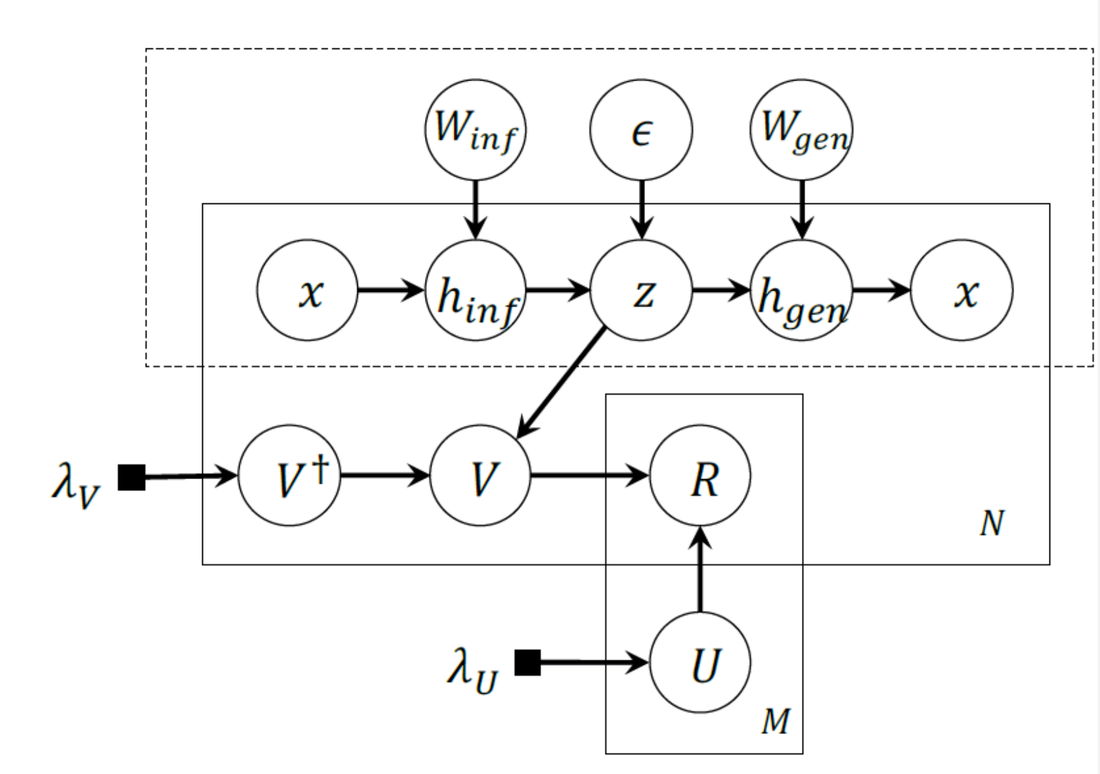
(a) *ML-20M*	Recall@20	Recall@50	NDCG@100
popularity	0.162	0.235	0.191
EASE^R	0.391	0.521	0.420
EASE$^R \geq 0$	0.373	0.499	0.402
results reproduced from [13]:			
SLIM	0.370	0.495	0.401
WMF	0.360	0.498	0.386
CDAE	0.391	0.523	0.418
MULT-VAE ^{PR}	0.395	0.537	0.426
MULT-DAE	0.387	0.524	0.419
(c) *MSD*			
popularity	0.043	0.068	0.058
EASE^R	0.333	0.428	0.389
EASE$^R \geq 0$	0.324	0.418	0.379
results reproduced from [13]:			
SLIM	— did not finish in [13] —		
WMF	0.211	0.312	0.257
CDAE	0.188	0.283	0.237
MULT-VAE ^{PR}	0.266	0.364	0.316
MULT-DAE	0.266	0.363	0.313

Steck, H. (2019, May). Embarrassingly shallow autoencoders for spar 257).

CVAE

- CVAE: Collaborative Variational Autoencoder
- Combina Collaborative Filtering y Content-Based
- Aborda el problema de cold start y sparsity.
- Aprende la representación del usuario, de las interacciones del ítem y del contenido del ítem.

$$\hat{R}_{ij} = u_i^\top (v_j^\dagger + z_j)$$



Li, X., & She, J. (2017, August). Collaborative variational autoencoder for recommender systems. In Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining (pp. 305-314).

MULTVAE

- Utiliza la matriz de interacción usuario-item
- El encoder aprende una representación latente del usuario u
- El decoder toma la representación y genera una distribución multinomial sobre los ítems

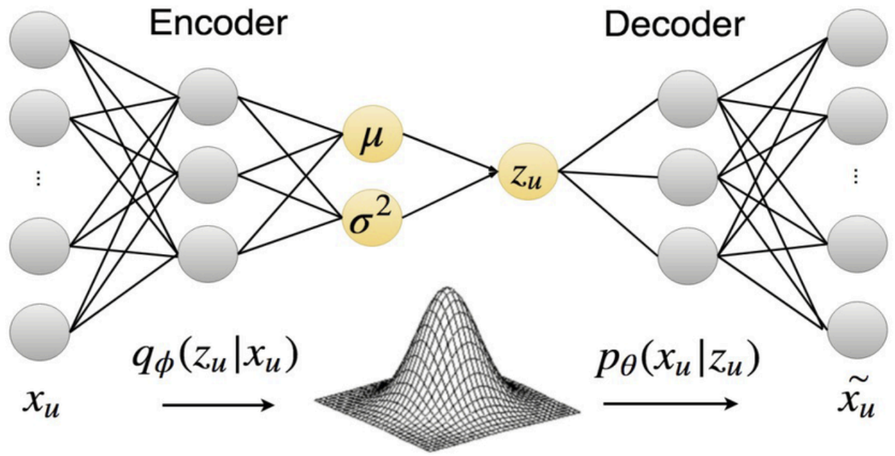


Figure 1: Variational Autoencoders for collaborative filtering. For each user: (1) a bag-of-items vector x_u is provided as input to the encoder; (2) a latent user vector z_u is sampled from a Gaussian distribution with parameters specified by the encoder; (3) a new bag-of-items vector $\tilde{x}_u$ is reconstructed via the decoder.

Liang, D., Krishnan, R. G., Hoffman, M. D., & Jebara, T. (2018, April). Variational autoencoders for collaborative filtering. In Proceedings of the 2018 world wide web conference (pp. 689-698).

DEEP FM

- Es un modelo híbrido para predecir CTR (Click-Through Rate)
- Usa Factorization Machines para aprender relaciones de bajo orden.
- Usa una Red Neuronal para aprender relaciones de mayor orden.
- Recibe como entrada un vector de features del usuario con la interacción y el contexto.
- Retorna la probabilidad de que el usuario haga click en un ítem

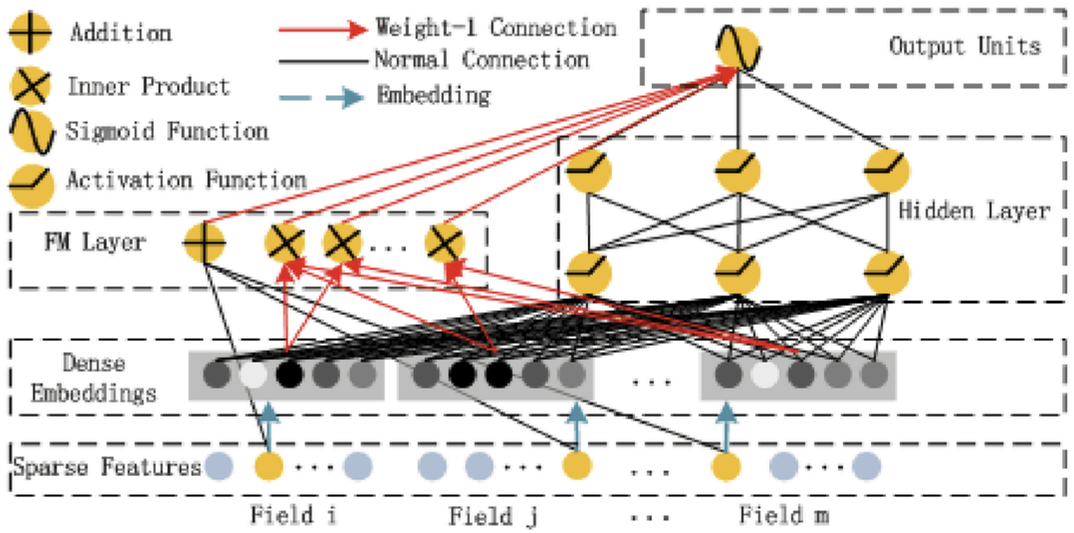


Figure 1: Wide & deep architecture of DeepFM. The wide and deep component share the same input raw feature vector, which enables DeepFM to learn low- and high-order feature interactions simultaneously from the input raw features.

Guo, H., Tang, R., Ye, Y., Li, Z., & He, X. (2017). DeepFM: a factorization-machine based neural network for CTR prediction. arXiv preprint arXiv:1703.04247.

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